

is still an expanding topic. Dr Reger adds, "It is still unclear what the optimal depth of a deep learning net should be, or what the ideal component functions in the targeted approximation might be."

Upgrading neural networks

It is no doubt that neural networks can identify and recognise patterns and do a lot of other interesting stuff. However, when we talk about real-time image analysis from multiple angles and lack of content in the frame, going beyond the capabilities of neural networks is required. One of the most stated advancement in this regard has been convolutional neural networks (CNNs). Self-driving cars, auto-tagging of friends in pictures, facial-security features, gesture recognition and automatic number plate recognition are some of the areas that benefit from CNNs.

Broadly, a CNN consists of three steps that can be divided into more steps. Starting with convolution, we go back to our engineering days, when we convolved a filter with input function to get some output. Padding was then added to make this result similar in size to the original image. You know the basics.

Pooling is done to reduce the size of data, and this makes it easier to sift through. Then, the networks use fully-connected layers where each pixel is considered as a separate neuron similar to a regular neural network.

Batch normalisation was another step in the process, which has become outdated with use. Better results over CNNs result in similar implementation as well.

Where we have seen CNN before. Video analysis, image recognition and drug discovery are some of the common uses of CNNs around us. However, the one that brought CNNs to light would be AlphaGo programme by Google DeepMind. The

Chinese game of Go is a complex board game requiring intuition, creative and strategic thinking. It has been a major subject in artificial intelligence (AI) research. In March 2016, Google's AlphaGo beat Lee Sedol, a South Korean professional Go player of '9 dan' rank by 4-1. And here we thought recognising and following a person in a crowd was amazing for a neural network.

Lesser-known image analytics

Image analysis can also be done using many other techniques. These can be as simple as a binary tree with a simple true-false decision, or as complex as structured prediction. Decision tree, for example, is used by Microsoft Structured Query Language (SQL) server. Never thought we would be using analytics in SQL, did we? But when you think about how data is fetched in SQL, it makes a lot more sense.

Decision tree is a predictive modelling approach used in statistics, data mining and machine learning. An analogy for easier understanding would be the binary tree. Some of the popular implementations are IBM SPSS Modeler, RapidMiner, Microsoft SQL Server, MATLAB and the programming language R, among others.

In R, decision tree uses a complexity parameter to measure trade-offs between model complexity and accuracy on training set. A smaller complexity parameter leads to a bigger tree, and vice versa. If you have

a smaller tree, it means that the model did not capture underlying trends properly, and the tree needs to be re-examined.

Divide and conquer to analyse. Cluster analysis, on the other hand, categorises objects in test data into different groups or clusters. Test data can be grouped into clusters based on any number of parameters, resulting in multiple algorithms in cluster analysis. Clusters are formed based on your algorithm, making it a multi-objective optimisation problem.

Formation of clusters, however, is not successful with the current data analysis needs. With the introduction of the Internet of Things (IoT), data that needs to be analysed has reached much higher volume. Many methods fail due to the curse of dimensionality, resulting in many parameters being left out while optimising the algorithm.

If you can remove some parameters in clustering, removing it altogether is not a stretch. Dimension reduction is the process of reducing the number of random variables being considered. The set of principal variables is then processed through feature selection and extraction. Data is classified by filters based on the features to be added or removed while building the model based on prediction errors.

Feature extraction then transforms this high-dimensional data into fewer dimensions through principal component analysis, among other techniques. Application areas include neuroscience and searching on live video streams, among others, as removal of multi-collinearity improves the learning system.

Odd man out algorithms. Used mostly in data mining, anomaly detection is another interesting algorithm. Anomalies, also referred to as outliers, novelties, noise, deviations and exceptions, can be a fancier version of the odd man out. Typical anomalies in-



The Scream by Edvard Munch on the left, and the horrifying scream after exaggeration of patterns on the right